

ARE OSTERSUND, Sweden

WMO: 022260

Lat: 63.197N

Lon: 14.486E

Elev: 376

StdP: 96.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.0	(c) -18.8	(d) -24.3	(e) 0.4	(f) -21.8	(g) -21.1	(h) 0.6	(i) -18.3	(j) 12.9	(k) 0.2	(l) 11.5	(m) -1.2	(n) 1.3	(o) 290	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.7	(c) 24.2	(d) 15.6	(e) 22.3	(f) 14.9	(g) 20.3	(h) 14.2	(i) 17.1	(j) 21.9	(k) 16.1	(l) 20.3	(m) 15.1	(n) 19.0	(o) 3.6	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.2	11.3	18.5	14.2	10.6	17.6	13.1	9.9	16.9	49.3	21.7	46.2	20.4	43.3	19.0	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.7	(b) 9.3	(c) 8.3	(e) -25.0	(f) 26.7	(g) 3.4	(h) 2.0	(i) -27.5	(j) 28.2	(k) -29.4	(l) 29.4	(m) -31.4	(n) 30.5	(o) -33.8	(p) 32.0		
(4) 10.7	(5) 9.3	(6) 8.3	(7) -25.2	(8) 18.5	(9) 3.4	(10) 1.5	(11) -27.6	(12) 19.6	(13) -29.6	(14) 20.5	(15) -31.5	(16) 21.3	(17) -33.9	(18) 22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-6.1	-5.9	-2.6	2.5	7.4	11.6	14.8	13.6	9.5	3.9	-1.0	-4.2		
	DBStd	8.55	6.08	6.27	4.75	3.59	4.05	3.50	3.42	3.15	3.01	3.52	4.26	5.82		
	HDD10.0	2693	499	444	391	224	103	21	2	4	44	191	331	440		
	HDD18.3	5363	757	677	649	474	339	203	121	151	266	448	581	698		
	CDD10.0	389	0	0	0	1	23	69	150	116	29	2	0	0		
	CDD18.3	16	0	0	0	0	0	2	11	4	0	0	0	0		
	CDH23.3	96	0	0	0	0	2	13	62	18	0	0	0	0		
	CDH26.7	7	0	0	0	0	0	0	7	1	0	0	0	0		
(14) Wind	WSAvg	3.9	3.9	4.0	4.0	3.6	3.9	4.1	3.7	3.7	4.0	4.2	4.0	3.8		
(15) Precipitation	PrecAvg	486	27	21	22	27	35	60	81	66	52	38	33	31		
	PrecMax	761	58	79	50	75	100	131	215	170	167	162	68	83		
	PrecMin	303	3	0	0	4	0	3	0	11	0	10	5	4		
	PrecStd	95	14	14	11	17	21	34	46	34	31	26	15	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.1	8.5	16.4	22.9	25.1	27.6	25.6	19.4	14.2	9.9	6.9		
		MCWB	3.3	3.4	4.2	9.0	13.4	14.3	17.5	16.5	14.0	10.6	7.8	5.1		
	2%	DB	3.9	4.2	6.0	13.0	20.0	22.7	25.0	23.0	17.1	11.6	7.7	5.1		
		MCWB	2.3	2.5	3.2	6.8	11.9	14.1	16.6	15.7	12.8	9.2	6.2	3.4		
	5%	DB	2.8	3.1	4.7	10.3	17.1	20.0	23.0	20.9	15.5	10.1	5.7	3.8		
		MCWB	1.5	1.6	2.3	5.8	10.4	13.2	15.5	15.0	12.0	8.1	4.3	2.5		
	10%	DB	1.4	1.9	3.5	8.1	14.2	17.9	20.9	18.9	14.0	8.9	4.2	2.6		
		MCWB	0.4	0.5	1.4	4.3	9.1	12.1	14.9	14.4	11.2	7.3	3.1	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.8	4.7	9.2	14.2	16.6	19.0	18.2	14.9	11.5	7.9	5.2		
		MCDB	4.9	5.4	7.2	15.5	20.7	21.6	24.5	22.6	17.8	13.6	9.7	6.6		
	2%	WB	2.5	2.6	3.5	7.4	12.6	14.8	17.5	16.9	13.6	9.8	6.1	3.6		
		MCDB	3.6	4.0	5.6	11.6	18.4	20.2	22.8	21.0	16.3	11.3	7.5	4.8		
	5%	WB	1.5	1.7	2.6	6.1	11.1	13.7	16.5	15.9	12.5	8.5	4.4	2.5		
		MCDB	2.5	2.9	4.4	9.5	16.3	18.8	21.4	19.4	14.9	9.9	5.3	3.6		
	10%	WB	0.4	0.5	1.5	4.9	9.5	12.6	15.5	14.9	11.5	7.3	3.2	1.4		
		MCDB	1.3	1.7	3.3	7.7	13.9	17.1	19.8	18.2	13.6	8.8	4.1	2.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.7	6.5	7.3	8.0	7.6	7.7	7.0	5.8	4.4	3.9	4.8		
		MCDBR	5.4	5.2	6.4	10.9	12.2	10.9	10.6	9.7	8.0	5.9	5.1	5.3		
	5% WB	MCWBR	4.6	4.5	4.7	6.2	6.2	5.3	4.5	4.6	4.7	4.4	4.5	4.5		
		MCWBR	5.3	5.1	6.1	9.5	10.8	9.6	9.4	8.3	7.1	5.5	4.8	5.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.203	0.264	0.293	0.326	0.332	0.328	0.342	0.332	0.303	0.287	0.213	0.346		
		taud	2.049	2.255	2.332	2.423	2.479	2.516	2.490	2.509	2.575	2.483	2.126	2.790		
	Ebn at Noon	Ebn at Noon	446	690	815	861	885	895	870	848	813	672	426	278		
		Edh at Noon	29	57	79	91	94	92	92	83	64	47	26	18		
(44) All-Sky Solar Radiation	RadAvg	0.18	0.77	2.14	3.79	4.65	5.32	4.92	3.69	2.22	0.96	0.26	0.09	0.01		
	RadStd	0.02	0.10	0.21	0.32	0.46	0.39	0.55	0.31	0.23	0.14	0.04	0.01	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(47) Regional (48 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(46) Station Only	(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-123	N/A	N/A	N/A

Nomenclature: See separate page